

LECTURE 1

Applied AI in Business - from opportunity to decision

Applied AI in Business

Armando Vieira

Lecture 1

A fluent answer is not a business outcome

AI becomes valuable only when it improves a real decision or performs a useful action safely.

The question for this course

How do we move from an impressive model demonstration to a reliable organizational capability?

The course builds decision systems

Analytical capability

- Statistical thinking
- Visualization and experimentation
- Forecasting, scoring and segmentation
- LLM-based analysis

Operational capability

- Agents that use tools
- Controlled workflows and approvals
- Evaluation and observability
- Governance and organizational adoption

Course promise

Become an analytics translator who connects evidence, technology, business action and organizational implementation.

By the end, you should be able to...

- Identify valuable AI opportunities and reject weak ones.
- Translate an organizational problem into a precise decision task.
- Combine data, predictive models, LLMs, tools and business rules.
- Build bounded agents with explicit permissions and human oversight.
- Evaluate value, reliability, cost, risk and organizational readiness.
- Communicate evidence and trade-offs to technical and business stakeholders.

The course moves from evidence to action

Discover → **Model** → **Reason** → **Act** → **Prove**

- Discover** Frame the decision, assess readiness and establish trustworthy evidence.
- Model** Add forecasting, scoring, segmentation or experimentation where useful.
- Reason** Use LLMs for documents, structured outputs and evidence-grounded analysis.
- Act** Build agents and harnesses that control tools, state and permissions.
- Prove** Evaluate, govern, pilot and communicate the complete system.

Construction and critique drive learning

During class

Short concept blocks, demonstrations, guided development, discussion and failure analysis.

Between classes

Improve one cumulative project using evidence and feedback from each lecture.

The standard is not “it ran”

- Does it solve a relevant problem?
- Is the evidence credible?
- Does it fail safely?
- Can others inspect and reproduce it?
- Will people adopt it?

The stack supports accountable building

Development	Python, notebooks, Git and Cursor as the primary AI-assisted IDE.
Cloud fallback	Google Colab when a local setup is unavailable.
Analytics	pandas, visualization, statistics and machine-learning libraries.
Generative AI	An institution-approved LLM endpoint or local model.
Data and tools	CSV/SQL data, documents, APIs and simulated business systems.
Reliability	Tests, schemas, evaluation datasets, traces and approval gates.

AI-assisted coding retains human responsibility

Use AI to accelerate

- Code generation and explanation
- Refactoring and documentation
- Test creation and debugging
- Repository navigation

Humans remain responsible for

- Data meaning and privacy
- Statistical validity
- Correctness and security
- Reproducibility and decisions

Practice rule

Generated code must be understood, tested, reviewable and reproducible.

Applied AI performs five kinds of work

Predict	Estimate an unknown quantity or future event.	Demand, churn, risk, revenue
Generate	Produce or transform unstructured content.	Reports, summaries, drafts
Recommend	Rank options or propose a next action.	Leads, products, interventions
Optimize	Allocate scarce resources under constraints.	Inventory, pricing, scheduling
Act	Use tools to change a system or advance a process.	Update records, send drafts, open cases

A business system may combine several of these capabilities.

Match the technology to the problem

Rules

Best for stable, explicit processes. Brittle when situations vary.

Statistical analysis

Best for explanation and uncertainty. Usually informs rather than acts.

Machine learning

Best for repeated prediction. Requires labels, validation and monitoring.

Design principle

Choose the least complex approach that can reliably produce the required business outcome.

Language and action introduce new options

Large language models

Useful for language, documents, extraction and flexible generation.

Constraint: outputs are probabilistic and require task-specific evaluation.

Agents

Useful for multi-step tasks that require decisions, tools and adaptation.

Constraint: autonomy adds operational risk and system complexity.

Not every chatbot is an agent

System	What it does	Who chooses?
LLM call	Returns one response	User
Copilot	Assists a person	Person
Workflow	Runs predefined steps	Designer
Agent	Chooses tools and actions during the task	Model within policy
Agent team	Coordinates specialized agents	Orchestrator

An agent operates inside a control loop

Goal → **Observe** → **Decide** → **Use tools** → **Verify**

The agent needs

- Context and state
- Available actions
- Tool interfaces
- Stopping conditions

The organization needs

- Permissions and policies
- Validation and approvals
- Logs and accountability
- Recovery procedures

Autonomy is a design choice

1. **Inform:** retrieve and summarize relevant information.
2. **Draft:** prepare an output for human review.
3. **Recommend:** propose a decision and explain the evidence.
4. **Act with approval:** prepare tool actions that a person authorizes.
5. **Bounded autonomy:** act independently within explicit limits.

The right level depends on consequence

Higher impact, ambiguity or irreversibility demands stronger constraints and human control.

The harness makes agents operational

Orchestration

Allowed steps, order and termination.

State and memory

Task progress and durable information.

Tool registry

Available actions and validated interfaces.

Policy layer

Permissions, privacy and approvals.

The model supplies capability; the harness determines what is allowed, remembered, observed and audited.

Evaluation

Output, action and task-level checks.

Observability

Traces, latency, cost and outcomes.

Recovery

Timeouts, duplicates, partial failures and rollback.

Begin with the decision, not the model

Weak formulation

“We should use an LLM and build an agent.”

Decision formulation

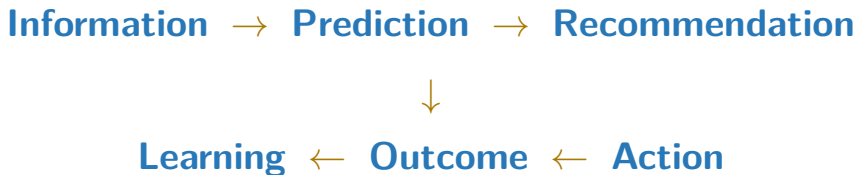
“Help a named decision owner choose an action, at a defined moment, under measurable constraints.”

Technology follows the decision architecture.

A decision specification answers seven questions

1. Who owns the decision and who is affected?
2. What action or choice can actually change?
3. When and how often is the decision made?
4. What information is available at that moment?
5. What constraints and approvals apply?
6. What are the costs of different errors and delays?
7. What baseline and outcome will define improvement?

Value requires action and learning



Where projects usually break

The model produces information, but no owner, workflow or feedback mechanism converts it into better outcomes.

Success must be measured at four levels

Level	Typical question	Illustrative measures
Model	Is the output technically adequate?	Error, calibration, groundedness
Workflow	Does the process work reliably?	Completion, latency, intervention, recovery
Adoption	Do people use and trust it appropriately?	Usage, override, satisfaction, training
Business	Does it improve the organizational outcome?	Revenue, cost, risk, service, capacity

A model metric is evidence, not the final objective

Technical improvement matters only when it changes workflow performance or business outcomes.

Errors have unequal consequences

Direct costs

- False positives and false negatives
- Human review and rework
- Latency and service interruption
- Model and infrastructure cost

Systemic costs

- Unfair or inconsistent treatment
- Privacy and security incidents
- Loss of trust and adoption
- Irreversible or duplicated actions

Design implication

Thresholds, approvals and autonomy should reflect business loss, not only average model accuracy.

Sometimes the correct AI solution is no AI

- The process is poorly defined or changes before outcomes can be measured.
- A simple rule, form, dashboard or process redesign solves the problem.
- The organization lacks reliable data or a legitimate basis for using it.
- Errors are consequential, difficult to detect and hard to reverse.
- No one owns the decision, the workflow or the resulting actions.
- Expected value does not justify development, operation and governance costs.

Restraint is part of AI competence.

Capability without readiness creates prototypes

Process

Is the task stable, understood and owned?

Data

Are definitions, quality, access and lineage adequate?

Technology

Can the system integrate, scale and recover?

Governance

Are permissions, accountability and controls explicit?

People

Do users have incentives, skills and trust to adopt it?

Economics

Is expected value greater than total operating cost and risk?

Prioritize opportunities on four dimensions

1. Business value

Frequency, scale, delay, error cost and strategic relevance.

2. Feasibility

Data, integration, technical capability and evaluation access.

3. Risk

Consequence, privacy, security, fairness and reversibility.

4. Readiness

Ownership, process maturity, governance, skills and adoption.

A strong first project

High value, observable outcomes, bounded actions, available data and manageable failure consequences.

Five ideas to carry forward

1. Start with a decision and an action, not a fashionable technology.
2. Use the simplest method capable of producing the required outcome.
3. Treat agent autonomy as a bounded design variable.
4. Evaluate the model, workflow, adoption and business result separately.
5. Organizational readiness is part of system performance.

Now apply the framework

First examine one complete business case; then design and stress-test your own AI opportunity.

WORKED EXAMPLE

A bounded agent for sales lead prioritization

From an overloaded queue to a controlled decision workflow

Illustrative scenario - not based on a specific company

The bottleneck is a repeated decision

Current process

- Leads enter the CRM from several channels.
- Representatives inspect records and research each company.
- Each representative decides priority, next action and message.
- Outcomes are recorded inconsistently.

Operational problems

- Limited time and uneven research depth
- Inconsistent prioritization criteria
- Missing or stale information
- Weak feedback from outcomes

Candidate decision: Which leads deserve attention now, and what is the safest useful next action?

A decision specification makes the task testable

Owner	Sales representative, supported by sales operations.
Decision	Prioritize the lead and choose the next approved action.
Moment	When a new lead arrives and when the daily queue is refreshed.
Inputs	CRM history, engagement, company fit, approved sources and team capacity.
Possible actions	Research, request missing data, draft outreach, defer or escalate.
Constraints	No autonomous external contact in the pilot; evidence and permissions required.
Baseline	Time per lead, queue delay, conversion among contacted leads and manual overrides.

Several solutions are possible

Option	Contribution	Remaining gap
Rules + workflow	Enforce constraints and run stable steps.	Cannot rank ambiguous cases well.
Predictive score	Estimates conversion or expected value.	Does not explain context or act.
LLM copilot	Summarizes evidence and drafts messages.	Depends on the user to gather inputs.
Bounded agent	Selects approved tools and handles variation.	Requires stronger controls and evaluation.

Design choice: Use rules and prediction for structure, an LLM for language, and an agent only where the next step varies.

The hybrid separates capability from control

Lead request → Retrieve + validate → Score → Summarize



Learn from outcome ← Log action ← Human approval ← Propose

Probabilistic capability

Scoring, summarization and recommendation.

Deterministic control

Schemas, rules, permissions, approvals and logs.

Read-only tools can run automatically

Tool	Pilot permission	Control
Read CRM record	Allowed	Access control and field filtering
Query approved sources	Allowed	Domain allow-list and source capture
Calculate lead score	Allowed	Versioned model and input validation
Draft recommendation	Allowed	Structured output and evidence links

Automatic does not mean uncontrolled

Every tool has a narrow schema, filtered data access, recorded evidence and a visible execution trace.

Consequential actions remain human-owned

Change CRM priority

The agent prepares the update; a representative reviews the evidence and confirms it.

Controls: preview, confirmation, version check and audit log.

Pilot boundary

Expand autonomy only when evidence shows that the action is reliable, reversible and governable.

Send an external message

The agent prepares a draft; a representative sends it through the existing system.

Controls: no messaging credential is available to the agent.

Evaluation covers the complete system

Technical evidence

- Ranking and calibration quality
- Evidence coverage and groundedness
- Tool-call and schema accuracy
- Normal and edge-case completion

Guardrails

- Zero unauthorized external actions
- No unsupported factual claims

Operational evidence

- Review time and queue delay
- Human override and escalation
- Failure recovery and duplicate prevention

Business and adoption

- Capacity and conversion outcomes
- Appropriate use and user trust

Information failures require abstention

Failure	Required response
Missing CRM fields	Ask for information or abstain; never invent values.
Stale company data	Show timestamp, reduce confidence and request review.
Unsupported claim	Require evidence links or remove the claim.
Manipulative retrieved text	Treat external text as data, not instructions.

Default behavior

When evidence is insufficient or conflicting, the system should expose uncertainty and return control to a person.

Operational failures require safe recovery

Failure	Required response
Duplicate update	Use an idempotency key and inspect current state.
Unstable subgroup performance	Monitor segments and suspend affected use.
Revoked permission or tool outage	Stop safely, preserve state and escalate.
Partial workflow failure	Record completed steps and resume only after validation.

Default behavior

A failed action must not silently become a duplicate, an unauthorized workaround or a lost audit trail.

Start with a controlled pilot

1. Run in **shadow mode** and compare recommendations with current decisions.
2. Limit use to a small team, approved data and reversible actions.
3. Require human approval for CRM changes and all external contact.
4. Record inputs, recommendations, evidence, overrides, failures and outcomes.
5. Compare workflow, adoption and business results with the baseline.
6. Expand autonomy only after explicit evidence and governance review.

Pilot decision

Proceed only if the system reduces decision effort without degrading quality, control or trust.

Risks of AI implementation

Key risk areas to address

- **Accuracy & Hallucination** – Models generate plausible but false information with no built-in fact-checking
- **Bias & Fairness** – Training data reflects historical inequities, which AI can entrench and automate
- **Privacy & Security** – Models may expose sensitive training data or be vulnerable to prompt injection attacks
- **Explainability** – Black-box decision-making makes it difficult to audit, contest, or trust outcomes
- **Regulatory & Legal** – Non-compliance with GDPR, EU AI Act, or sector-specific rules carries significant liability

The silent crisis of AI adoption

Loss of organizational control

- **Disempowerment** — Decisions are made by algorithms, not people
- **Incomprehension** — No single expert understands the full system
- **Entropy** — Systems degrade faster than we can document them
- **Unmanageability** — Traditional risk frameworks simply do not apply

WORKSHOP

Design an AI opportunity that deserves a pilot

Frame the decision, bound the system and define the evidence

Teams of 3-4 | 55-minute design sprint | 20-minute peer review

Your output is a one-page opportunity brief

Choose one recurring process

- Sales or marketing
- Finance or risk
- Operations or supply
- Customer service
- Internal knowledge work

Submit one coherent design

- Current workflow and baseline
- Decision and proposed intervention
- Data, tools and autonomy boundary
- Metrics, risks and controls
- Smallest credible pilot

Do not begin with a technology

Begin with a costly, slow, inconsistent or information-heavy decision that an identifiable person owns.

The canvas connects four layers

1. Business decision

Problem, owner, affected users, action, frequency, baseline and expected value.

2. AI intervention

Capability, data, rules, models, LLM functions, tools and human role.

3. Evidence and control

Metrics, evaluation cases, error costs, permissions, approvals and recovery.

4. Organizational path

Readiness, ownership, adoption, pilot scope and go/no-go criteria.

Every technical component must connect to a decision, control or measurable outcome.

Step 1 - Map the current decision

1. Name the decision owner and affected stakeholders.
2. Describe the trigger, available information and possible actions.
3. Identify delay, inconsistency, error, rework or missed opportunity.
4. Define the current baseline using observable measures.

Challenge question

What will somebody do differently if the system works?

Time

10 minutes

Step 2 - Design the smallest useful intervention

1. Select the required capability: predict, generate, recommend, optimize or act.
2. Decide which steps are rules, models, LLM tasks or human decisions.
3. List only the data and tools required for the task.
4. Choose the lowest autonomy level that creates value.

Challenge question

Where is agent autonomy genuinely necessary?

Time

15 minutes

Step 3 - Define evidence and failure boundaries

1. Define model, workflow, adoption and business metrics.
2. Identify the three errors with the greatest consequences.
3. Specify permissions, approvals, stopping and escalation.
4. Create at least one normal, edge and adversarial test.

Challenge question

What evidence would persuade a skeptical decision owner?

Time

15 minutes

Step 4 - Make the pilot organizationally credible

1. Assess process, data, technology, governance and people readiness.
2. Define a limited user group, duration and action boundary.
3. Name the operational owner and human approval role.
4. State the evidence for proceeding, redesigning or stopping.

Challenge question

What is the smallest test that could invalidate your idea?

Time

10 minutes + 5 minutes to finalize

Peer review should try to break the proposal

Review another team's design

- Is the decision and owner precise?
- Is AI necessary and supported by credible evidence?
- What happens when data, tools or the model fail?
- Are autonomy and permissions proportional to consequence?
- Is the pilot small enough to control and strong enough to teach?

Review protocol

5 min presentation
5 min red-team questions
5 min written feedback
Switch teams

Close with a 90-second pilot case

1. **Problem:** Which decision is currently costly, slow or inconsistent?
2. **Intervention:** What will the system predict, generate, recommend or do?
3. **Control:** What remains deterministic or human-owned?
4. **Evidence:** Which measures and tests determine whether it works?
5. **Pilot:** What is the smallest safe deployment that can teach us something?

Exit ticket

Write one assumption most likely to destroy the business value of your proposal and one observation that would reveal it early.

The lesson ends where implementation begins

A strong AI opportunity connects a real decision, a bounded intervention, credible evidence and an organization ready to act.

Next lecture

Data, statistical thinking, visualization and responsible AI-assisted development.